

eROSITA × DESI × AION-flow — Experiment Pipeline

Living document — keep in sync with the actual pipeline. Last updated: 2026-07-17. Render to PDF with python docs/render_pipeline_pdf.py (writes docs/pipeline.pdf).

1. Goal

Use a frozen astronomical foundation model (AION-1 base) as an information probe: predict eROSITA X-ray properties (broad-band flux, luminosity, hardness) and host stellar mass from non-X-ray DESI modalities (optical spectra, WISE, redshift, Legacy imaging) with a probabilistic normalizing-flow head. Report likelihood gain over a prior, not just point R^2 . This repo extends the PAI26 paper with (a) a cleaned crossmatch and (b) uncertainty-aware, per-target training.

2. Compute

- FASRC** (Harvard SLURM). Account `siag_lab`; partition `siag_gpu` (32× A100); smokes on `gpu_test`.
- Env: `module load Miniforge3/26.1.0-fasrc01 + conda env + cu124 torch wheel + pip install -e . polymathic-aion[torch] zuko safetensors`.
- Non-interactive access via SSH ControlMaster socket `~/ .ssh/cm-fasrc`.

3. Data sources & download

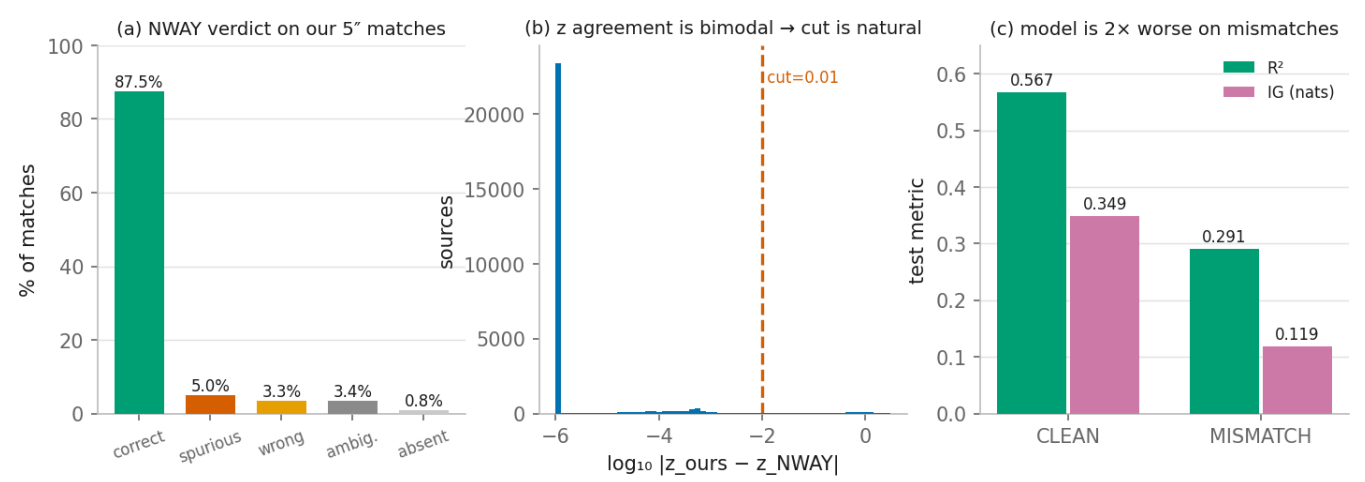
Modality	Source	Notes
X-ray catalog	SRG/eROSITA-DE DR1 / eRASS1 (Merloni+2024)	broad band <code>ML_FLUX_1</code> + sub-bands <code>P1–P4</code> rates/fluxes with asymmetric errors (LOWERR/UPERR). Delivered as the eROSITA×DESI match CSVs.
Optical spectra	DESI DR1 coadd spectra	per-healpix <code>coadd-{survey}-{program}-{healpix}.fits</code> , trimmed to matched targetids, compiled to <code>erosita_spectra_merged_32k.hdf5</code> .
Imaging	Legacy Survey DR10 griz cutouts	160×160 px <code>fits_pool/<targetid>.fits</code> via legacysurvey.org cutout service (~11 GB).
Redshift / galaxy props	DESI DR1 VAC	<code>z</code> , <code>logmstar</code> (photometric), emission lines, <code>spectype</code> (QSO/GALAXY/STAR).
Foundation model	<code>polymathic-ai/aion-base</code> (HuggingFace, ~95 MB)	frozen tokenizer + backbone.

All inputs are local; only `aion-base` is internet-bound. "Adding sources our match missed" would require new DESI DR1 spectrum downloads (deferred).

4. Crossmatch & cleaning

- Original match (existing):** naive **5" nearest-neighbor** (`crossmatch_radec.py`, `astropy`) between DESI target RA/Dec and the eROSITA catalog → 33,938 pairs → 30,441 unique DESI targets → **28,646 with a retrieved spectrum**.
- Quality problems found:** separations pile up near the 5" cut (median 3.08", 52% >3"); **14% of eROSITA sources match >1 DESI target** (26% of pairs in multi-match groups) → the same X-ray label duplicated onto several spectra.

- **Cleaning (NWAY):** validate against **Salvato et al. 2025 (A&A 704, A344)** — the eRASS1 Bayesian NWAY counterpart catalog — via VizieR J/A+A/704/A344/dr1ls10 (join on IAUName; compare their counterpart's z_{sp} to our matched DESI z). Per-targetid class: **87.5% correct, 4.6% spurious, 3.3% wrong, 3.4% ambiguous, 0.8% absent** → keep = correct filter (data/match_quality.csv).
- **Sanity checks (cleaning is sound, 3 independent angles):** (a) $|z_{\text{ours}} - z_{\text{NWAY}}|$ is **bimodal** — 88.7% $< 1e-4$ (same object), 3.0% > 0.1 (different), only 0.6% in the 0.01–0.1 valley → the z -cut is natural, not tuned; (b) mismatches show larger match separation (3.19" vs 3.06") but only weakly — separation can't discriminate at eROSITA's $\sim$ arcsec position errors, which is *why* prior-based NWAY is needed; (c) the trained model is **2× worse on mismatches** (R^2 0.29 vs 0.57) — independent confirmation the flags localize real failures.
- **Pipeline order (IMPORTANT):** clean → **dedup to the NWAY counterpart per targetid** → **re-derive the train/val/test split on the cleaned sample** (targetid-grouped 80/10/10, seed 42, leakage-safe) → stage. Splits must be redone *after* cleaning.



5. Targets & uncertainties

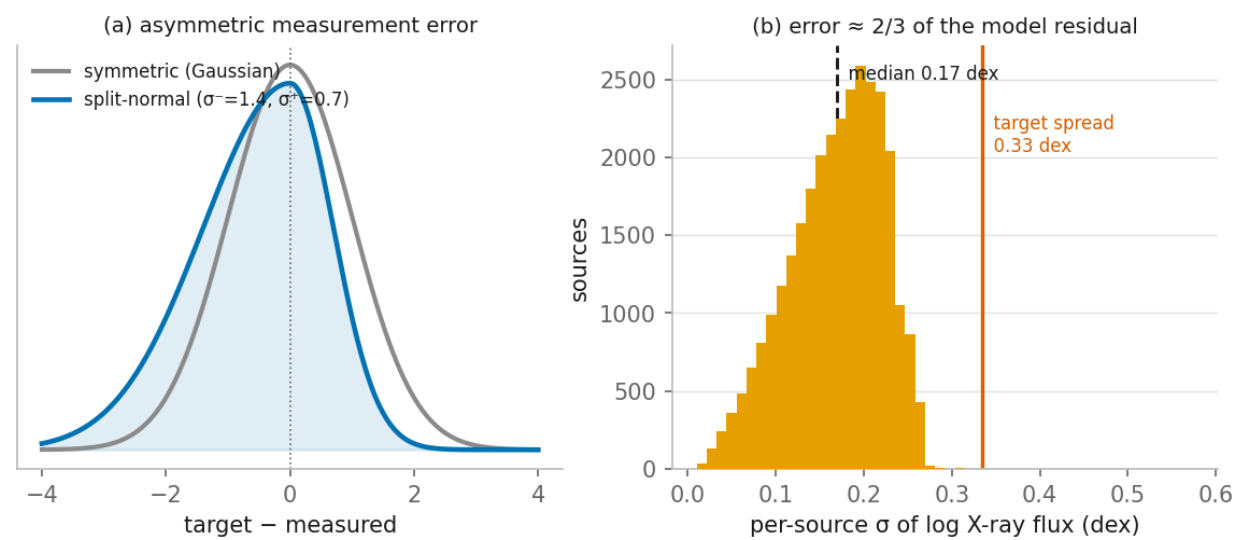
Four separate 1-D flow runs (data/targets_extra.csv carries the derived targets + errors, keyed by targetid):

Target	Source	Error model
log_ml_flux_1	$\log_{10}(\text{ML_FLUX_1})$	split-normal (two-piece Gaussian) from LOWERR/UPERR: $\text{sig_hi} = \log_{10}(1 + \text{UPERR}/F)$, $\text{sig_lo} = -\log_{10}(1 - \text{LOWERR}/F)$ (capped). Median ≈ 0.17 dex; 98% asymmetric, low side heavier.
log_lx	$\text{ML_FLUX_1} + z$ (Planck18)	same dex σ as flux.
logmstar	DESI VAC photometric mass	spectype systematic floor: 0.20 dex (GALAXY) / 0.30 dex (QSO). No per-source posterior σ available (PROVABGS covers only 0.4% of this AGN sample). Caveat: 82% of sample is QSO → mass AGN-contaminated; evaluate split by spectype.
hr32 (hardness)	$(P3 - P2) / (P3 + P2)$ band rates	modeled in arctanh space $u = \text{arctanh}(\text{HR})$, $\text{sig_u} = \text{sig_hr} / (1 - \text{HR}^2)$. Only $\sim 17\%$ have $S/N > 2$ ($\sigma \approx \text{signal}$) → IG-primary , R^2 secondary. HR43/T dropped.

Targets & error treatment (four separate V1 runs, all --error-mode convolve)

target	source	error model	mode	trainable
log_ml_flux_1	eROSITA ML_FLUX_1	split-normal (LO/HI), ~0.17 dex	convolve	full sample ✓
log_lx	flux + z (Planck18)	same dex σ as flux	convolve	full (z-dominated)
logmstar	DESI photometric mass	spectype floor 0.2/0.3 dex	convolve	clean for GALAXY; 82% QSO
hr32_u	eROSITA band rates (arctanh)	propagated σ_{HR}	convolve	IG-primary; ~17% S/N>2

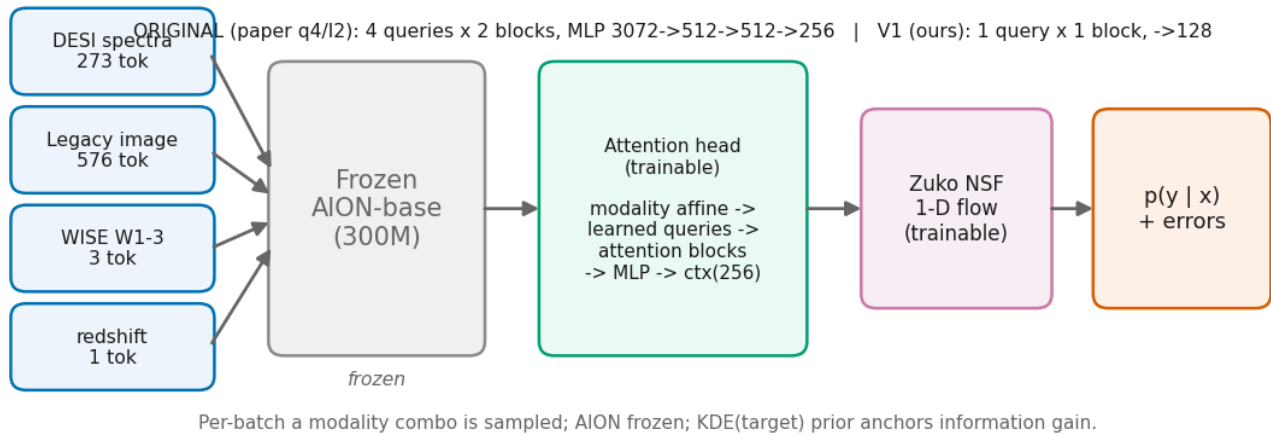
How errors enter the flow (split-normal convolution likelihood): $NLL_i = -\log \int p_{\text{flow}}(t \mid x_i) \cdot K(y_i \mid t; \text{sig_lo_i}, \text{sig_hi_i}) dt$, a fixed 1-D quadrature (~21–41 nodes over $\pm 4\sigma$), $\text{error_mode} \in \{\text{none}, \text{inject}, \text{convolve}\}$ (default **convolve** = deconvolves; **inject** broadens; **none** = paper behavior).



Measurement error is large relative to signal: for flux, $\sigma \approx 0.17$ dex is $\sim 2/3$ of the model's residual variance $\rightarrow$ the model sits near the measurement-noise floor, so error-aware training + a reported error floor matter.

Deferred error refinement: X-ray sub-band non-detections are **upper limits (censored)**, not missing/large-error; proper handling via a censored likelihood (ref 2022A&A...661A...3S). First pass uses split-normal σ on measured values.

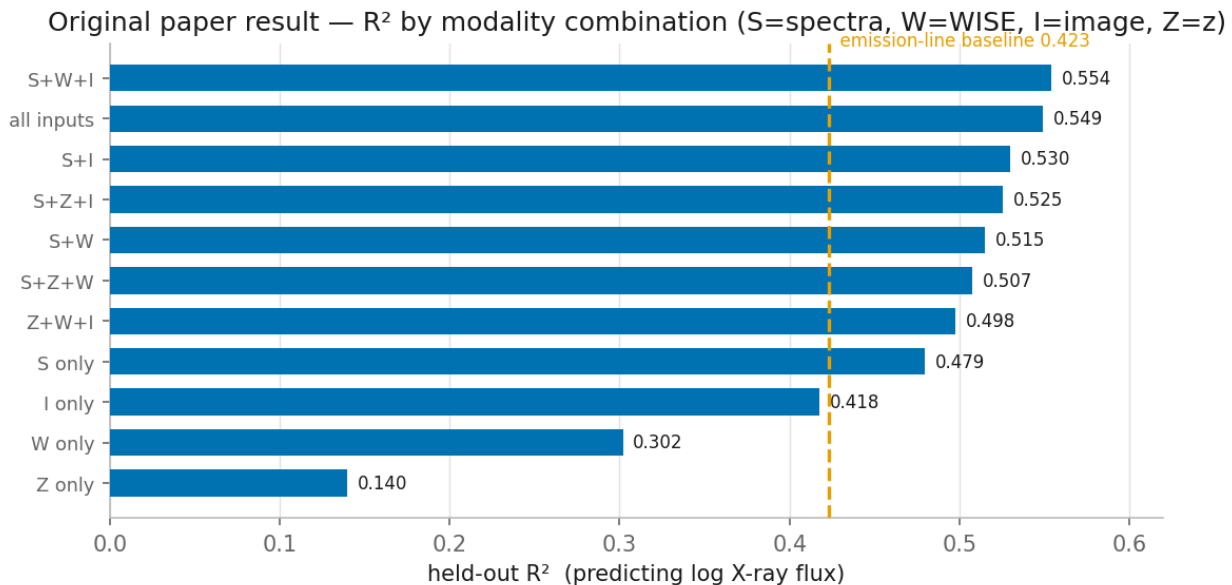
6. Model architecture



Frozen **AION-base** tokenizer + backbone → per-modality token sequences (spectra=273, image=576, WISE=3, z=1; missing modalities omitted). Trainable head + flow:

- **V1 minimal head** (AIONAttentionContext, parameterized): per-modality affine calibration → **1 learned query + 1 attention block (8 heads)** cross/self-attention → 768 → MLP (→128) → 256-d context. (~7.8M params vs the paper's 4-query/3072-d ~16.8M.)
- **Flow**: Zuko conditional **NSF**, 1-D target, 8 transforms, context 256; standardized target; **KDE (Scott)** prior as the IG reference.
- Per-batch a modality combo is sampled; AION frozen; AdamW; checkpoint best all-inputs val NLL.
- **V2 (later)**: replace frozen AION with a self-trained attention encoder over the raw spectrum.

Original paper result (q4/l2, epoch-13) — R^2 by modality combination; full spectra beat the emission-line baseline, all-modality is best:



7. Training & evaluation

- Four separate V1 runs (`--target {log_ml_flux_1, log_lx, logmstar, hr32}`), `--clean` (NWAY keep), `--error-mode convolve`, on `siag_gpu`; smoke on `gpu_test` with a `--limit` subset.
- **Metrics**: primary **IG / exp(IG)** vs KDE prior (esp. HR); secondary **R^2 + reduced- χ^2** using per-source σ + the measurement-error floor; **clean-vs-full delta**; per-spectype split for logmstar/HR.

8. Key findings so far

- Naive 5" match is ~8% wrong per NWAY; the paper's q4/l2 model is ~2× worse on the mismatches (R^2 0.29 vs 0.57); dropping them lifts test R^2 from the published 0.549 → 0.567 (no retraining).
- X-ray targets are measurement-noise-dominated (flux $\sigma \approx 0.17$ dex; HR $\sigma \approx$ its own spread) → uncertainty handling is essential.

9. Status

- ■ Repo cloned; V1 head coded+tested; FASRC scaffolding + **verify-before-push staging scripts** (`scripts/fasrc_preflight.sh`, `scripts/fasrc_stage_data.sh`).
- ■ Sidecars built: `data/match_quality.csv` (clean filter), `data/targets_extra.csv` (targets + split-normal/arctanh errors).
- ■ **Error-aware flow** implemented + tested (`split_normal_log_kernel`, `convolve_logprob`, `ConditionalNSFFlow.log_prob_convolved`; 14/14 tests, recovers analytic Gaussian convolution).
- ■ **Clean** → **dedup** → **re-split** implemented + tested (`build_clean_manifest`; 25,200 clean rows, exact 80/10/10, leakage-safe, deterministic).

- ■ **Full wiring done + validated locally:** `prepare-data --clean` cleans/dedups/re-splits + writes extra target & error columns; `AIONHDF5Dataset` emits per-source σ (10-tuple); `main.py --target {..., logmstar, hr32_u} / --error-mode {none, inject, convolve} / --clean`; `batch_nll` uses `log_prob_convolved`. 14/14 tests; a real `prepare-data --clean --limit 12` produced correct columns.
- ■ On FASRC: data staging ~76% (to `/n/netscratch/finkbeiner_lab/Lab/rditella/aionflow`) → then `code+env`, `prepare-data --clean`, and the V1 `log_ml_flux_1` smoke (`--error-mode convolve`) on `gpu_test`.
- ■■ `siag_lab` membership pending (Coldfront) → real `siag_gpu` runs after; `gpu_test` meanwhile.
- Known refinement: `hr32_u` piles at the `arctanh` clip for low-S/N HR (huge σ) — fine for `flux/lx/logmstar`; the HR run wants an S/N gate/clip.

10. Deferred / out of scope

SFR (needs DESI FastSpecFit VAC), Γ (=HR relabel), X-ray upper-limit censoring, full Salvato re-crossmatch of "the rest", V2 self-trained encoder, saliency/attribution.

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